Supplementary Material:
Minimal Diagnostic Principles of Life

Anonymous

Supplementary Material

S1 Mid-run and Invariance Protocol Extensions

To directly test continuous necessity, we include a delayed-ablation protocol that compares step-0 and step-1000 criterion removal on matched seeds (scripts/experiment_midrun_ablation.py; analysis via scripts/analyze_midrun.py). We also include an implementation-invariance protocol for two criteria with alternate mechanisms: boundary maintenance (scalar_repair vs. spatial_hull_feedback) and homeostasis (nn_regulator vs. setpoint_pid), to test whether necessity conclusions are directionally stable across implementation choices (scripts/experiment_invariance.py; scripts/analyze_invariance.py).

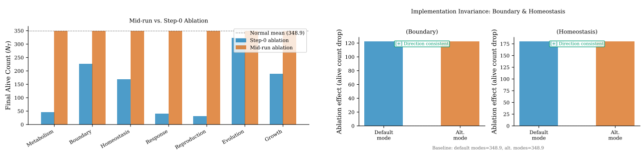

Figure S1: Protocol extensions for robustness. Left: step-0 vs. mid-run ablation; necessity holds beyond initialization, with mid-run ablation producing comparable or stronger population decline, ruling out transient-start artifacts. Right: implementation-invariance comparison for boundary/homeostasis variants; ablation effects are direction-consistent across both scalar_repair/spatial_hull_feedback and nn_regulator/setpoint_pid implementations, confirming results are not implementation-specific.

S2 Ecology Stressor Protocol Extension

Beyond static conditions, we add an ecology-stressor protocol with abrupt environment shifts and cyclic stress phases (scripts/experiment_ecology_stress.py) and a compact analysis pipeline (scripts/analyze_failure_pathways.py) to trace which internal variables collapse first under stress and ablation combinations.

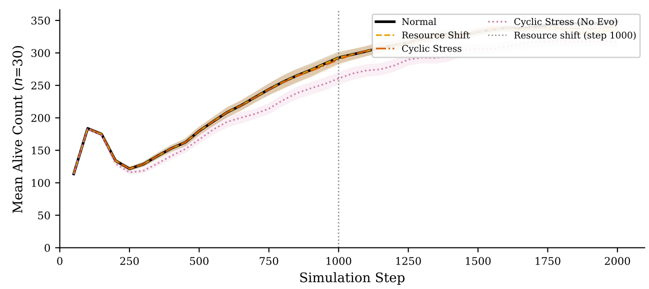

Figure S2: Ecology stressor conditions (n=30 seeds, 2,000 steps). Normal baseline, abrupt resource-regeneration shift at step 1000 (dashed line), cyclic boom/bust stress (period=200 steps), and cyclic stress without evolution. Mean alive count ±1 SEM. Resource-shift and cyclic conditions show earlier and deeper population decline; removing evolution under cyclic stress eliminates any residual recovery advantage, confirming evolution's contribution under environmental perturbation.

S3 Additional Validation

Phenotype clustering. PCA projection of per-seed organism traits (energy, waste, boundary integrity, genome diversity, generation count) reveals two emergent phenotypic clusters among evolved populations (k=2, silhouette=0.41; Figure S3). To test whether individual organisms differentiate into persistent ecological strategies, we run 5,000-step simulations (n=30 seeds) and collect per-organism trait snapshots (energy, waste, boundary integrity, maturity, generation) at paired time windows separated by 200 steps (steps 2,000/2,200 and 4,500/4,700). Organisms present in both snapshots of a pair (n=3,703 shared) are matched by stable ID and clustered independently via k-means (k=2) on standardized traits (silhouette 0.39–0.51; Figure S4). Despite meaningful within-snapshot cluster separation (two distinct strategies: high-energy/low-boundary vs. low-energy/high-boundary), the adjusted Rand index between paired windows is low (early-pair ARI=0.020, late-pair ARI=0.043), indicating that individual organisms transition between ecological roles even over short ($\sim$200-step) timescales. Using a pre-registered persistence threshold (stronger language allowed only when ARI $\geq$ 0.30), these values remain below the claim gate, so we report this as weak persistence rather than stable individual ecological differentiation. Cross-pair analysis (steps 2,000 to 4,700) finds near-zero organism overlap, confirming that median lifespan ($\sim$245 steps) prevents long-horizon individual persistence. An explicit long-horizon sensitivity run (10,000 steps, n=30) also remains below the pre-registered ARI claim gate (early-pair ARI=0.020, late-pair ARI=0.039; threshold $\geq$ 0.30), so the weak-persistence interpretation is robust to longer simulation horizons. A focused viability-threshold sweep (27 settings, 10 seeds each) shows the reproduction energy gate has the strongest effect on final alive counts, while varying the max-age cap over 15k–25k steps has negligible impact at the 2,000-step evaluation horizon. This suggests that ecological strategy differentiation in our system is a population-level rather than individual-level phenomenon: organisms continuously explore the trait space, with the population maintaining a stable distribution of strategies even as individuals transition between them, consistent with the moderate evolution effect (δ=0.39) observed in single-ablation experiments.

Spatial cohesion. Per-organism spatial cohesion (mean pairwise toroidal agent distance) under normal versus no-boundary conditions (n=30, 2,000 steps) confirms that boundary maintenance produces measurable spatial coherence, not merely scalar tracking (Figure S3, center).

Lineage structure. Parent–child tracking across all reproduction events yields multi-generational lineage trees (Figure S3, right), confirming genuine hereditary structure rather than random replacement.

S4 System Design Details

Rubric definition. Table S2 defines the five-level rubric used in Table 1 of the main paper.

Cellular organization: scalar vs. spatial representation. Boundary integrity $b \in [0, 1]$ tracks the functional consequence of swarm agent loss: as agents are lost, the organism's ability to maintain spatial coherence degrades, which is reflected in b decreasing. The spatial validation

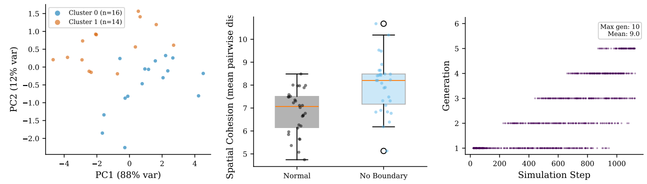

Figure S3: Additional validation. Left: phenotype clustering via PCA (k-means, silhouette-optimal k). Center: spatial cohesion under boundary ablation. Right: lineage phylogeny (generation depth vs. step).

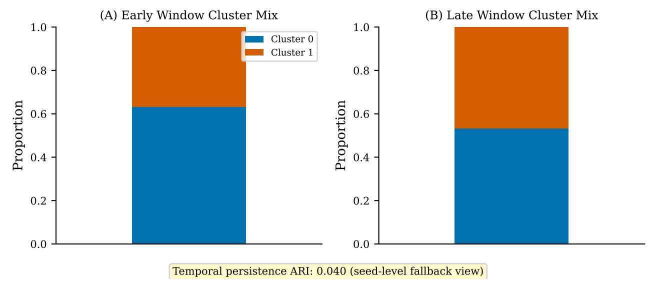

Figure S4: Per-organism ecological niche persistence. PCA projections of individual organism traits (energy, waste, boundary integrity, maturity, generation) at early (left) and late (right) time windows, colored by k-means cluster (k=2). Organisms are matched by stable ID across windows; the adjusted Rand index quantifies temporal persistence of cluster assignments.

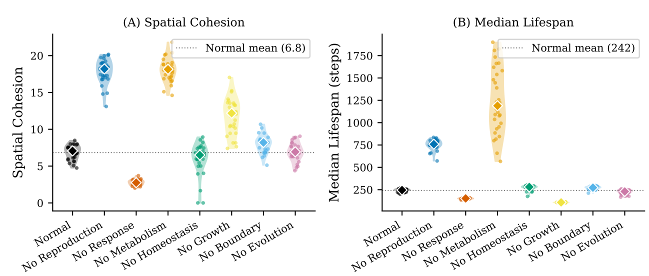

Figure S5: Criterion-orthogonal outcome measures (n=30 per condition). Left: spatial cohesion (mean pairwise agent distance); lower values indicate tighter spatial structure. Right: median organism lifespan per seed. Diamonds mark medians; dotted line shows normal baseline. Response ablation significantly increases cohesion (immobile organisms cluster) while decreasing lifespan.

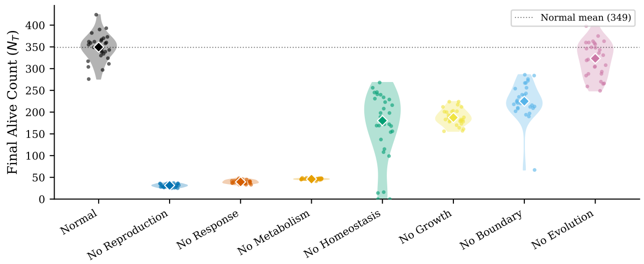

Figure S6: Per-seed distributions of final alive count (N_T) across all conditions (n=30 per condition). Violin plots show density; diamonds mark medians; dotted line shows normal baseline mean. Effects are consistent across seeds, not driven by outliers.

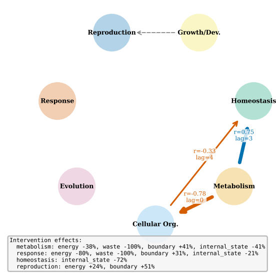

Figure S7: Criterion coupling graph. Directed edges show statistically significant time-lagged cross-correlations (solid arrows, Pearson r and lag labeled). Dashed arrows indicate design-level pathways (Table 2 in main paper).

Table S1: Intervention-based causal effects: percent change in criterion process rates when each criterion is ablated.

Ablated	Energy	Waste	Boundary	Int. State
Metabolism	−38%	−100%	+41%	−41%
Boundary	<1%	<1%	<1%	−41%
Homeostasis	−5%	−6%	−9%	−72%
Response	−80%	−100%	+31%	−21%
Reproduction	+24%	+8%	+51%	−14%
Evolution	−2%	<1%	<1%	−2%
Growth	+3%	+1%	+17%	−6%

Table S2: Five-level criterion implementation rubric.

Level	Definition
0	Criterion absent or purely cosmetic
1	Static parameter (no dynamics)
2	Single-process dynamics (no coupling)
3	Dynamic process with measurable degradation upon removal
4	Multi-process interaction with feedback coupling

uses spatial_cohesion_mean (mean pairwise agent distance), which serves as an independent spatial metric confirming that changes in b correspond to real spatial disaggregation (Figure S3, center). The scalar representation is a deliberate simplification: it enables clean ablation experiments (a single variable to track) while the spatial cohesion metric validates that the scalar captures genuine spatial effects.

Ablation implementation independence. Each ablation toggle skips a specific computational step without side effects on other criteria's code paths. For example, disabling metabolism freezes (e, r, w) values but does not alter the boundary decay equation, which still reads the current (frozen) energy value. This means criterion interactions emerge from shared state variables (energy, boundary) rather than implementation-level coupling. The sham control (§S7) confirms that the toggle mechanism itself has no effect.

Neural-network capacity. The fixed NN architecture (8→16 tanh→4 tanh; 212 weights) was chosen for computational efficiency rather than optimality. Whether NN capacity is a bottleneck for homeostasis or response quality remains an unexplored limitation; architecture search or larger networks might reveal stronger criterion interactions.

Architecture and runtime state. Each organism maintains: boundary integrity ($b \in [0, 1]$), metabolic state (energy e, resource r, waste w), internal state vector $\mathbf{s} \in \mathbb{R}^3$ for homeostatic regulation, a feedforward neural-network controller (8 inputs → 16 hidden tanh → 4 outputs tanh; 212 weights), a genetically encoded metabolic network, age, generation counter, and maturity level $m \in [0, 1]$.

Cellular organization. Boundary integrity decays each step: $\Delta b_{\text{decay}} = -r_b(1 + s_e(1 - e) + s_w w)$ (r_b=0.001, s_e=0.02, s_w=0.5). Repair is proportional to energy: $\Delta b_{\text{repair}} = r_r \cdot e \cdot (1 - s_p w)$ (r_r=0.05, s_p=0.4). Death occurs when $b < 0.1$.

Metabolism. Each organism's metabolic network has 2–4 catalytic nodes (16-float genome segment, sigmoid-decoded): node count = round($\sigma(g_0)\cdot 2+2$), catalytic efficiency = $\sigma(g_{2+i})\cdot 0.9+0.1$, edge existence $|g_j|>0.3$, conversion efficiency = $\sigma(g_{13})\cdot 0.7+0.3$. Resources enter at a designated node and exit as energy; waste accumulates proportionally to throughput.

Homeostasis. Internal state s_0 is actively maintained near 1.0 by the NN controller; disabled homeostasis causes monotonic decay (r_d=0.001/step), distinguishing active regulation from static parameterization.

Growth and development. Organisms progress through three developmental stages (juvenile, adolescent, adult) governed by a DevelopmentalProgram (8-float genome segment 3): maturation rate modifier (2^{g_0}, [0.25, 4.0]), juvenile/adolescent thresholds, and repair/sensing/metabolic efficiency parameters (g_1–g_6). Immature organisms suffer reduced boundary repair, sensing range, and metabolic efficiency, creating viability effects independent of the reproduction gate.

Reproduction. Division requires $e > e_{\min}$=0.7 and $b > b_{\min}$=0.5; parent pays energy cost c_r=0.3; offspring inherits a (possibly mutated) genome.

Evolution. Offspring genomes undergo point mutations (rate 0.02/gene, scale 0.15), reset mutations (rate 0.002), and scale mutations (rate 0.002, factor $\in [0.8, 1.2]$); all values clamped to $[-2, 2]$.

Genome encoding. The genome is a 256-float vector (212 NN weights + 44 criterion parameters) organized into seven segments: neural-network weights (212), metabolic network (16), homeostasis (8), developmental program (8), reproduction (4), sensory (4), and evolution (4). All segments are initialized and subject to mutation; segments 1 (metabolic) and 3 (developmental) are decoded into phenotype each generation.

Simulation parameters.

Parameter	Value
World	100×100 toroidal
Organisms	30 (25 agents/org)
Steps	2,000 (sample /50)
Resource	0.01/cell/step
Mutation	0.02/gene, s. 0.15
Cal. seeds	0–99
Test seeds	100–129 (n=30)

S5 Pairwise Ablation Analysis

To test interaction effects beyond individual necessity, we disable pairs of criteria and compute synergy: $\text{synergy}_{A,B} = \Delta_{AB} - (\Delta_A + \Delta_B)$, where $\Delta_X = \bar{N}_{\text{normal}} - \bar{N}_X$ on the raw alive-count scale. Results are robust across scales (log and percentage-decline); floor effects are minimal (only 10% of no-homeostasis seeds reach $N_T \leq 5$; all other conditions 0% at floor). Subadditivity thus reflects genuine shared failure pathways rather than mechanical floor compression.

The (metabolism, homeostasis) pair exemplifies the shared pathway: disabling metabolism eliminates energy production, starving boundary repair ($\Delta b_{\text{repair}} \propto e$); disabling homeostasis degrades adaptive behavior, accelerating waste accumulation and amplifying boundary decay via $s_w \cdot w$ in Δb_{decay}. Both routes converge on boundary failure, explaining why Δ_{AB}=298.1 falls below the additive expectation of 450.7.

Table S3: Pairwise ablation synergy scores (n=30 per condition, Graph metabolism). Negative synergy indicates sub-additive interaction. Baseline $\bar{x}$=348.9, consistent with Table 3 in the main paper.

Pair	Δ_{AB}	Exp.	Syn.	Ratio
(Metab, Homeo)	298.1	450.7	−152.6	0.66
(Metab, Resp.)	295.9	596.6	−300.7	0.50
(Reprod, Growth)	310.0	465.7	−155.7	0.67
(Bdry, Homeo)	111.4	261.5	−150.1	0.43
(Resp., Homeo)	303.8	456.7	−152.9	0.67
(Reprod, Evol)	310.0	330.0	−19.9	0.94

Exp. = Δ_A+Δ_B; Ratio = Δ_{AB}/Exp.

Growth–reproduction interaction. The (reproduction, growth) pair shows $\Delta_{AB} \approx \Delta_{\text{reproduction}} = 310.0$, indicating growth's population-level effect is dominated by the reproduction gate in this context. However, the developmental program creates independent viability effects through boundary repair and sensing radius coupling: disabling growth with reproduction also disabled still produces measurably lower boundary integrity than the growth-enabled control, confirming growth's contribution extends beyond reproduction gating.

S6 Evolution Timescale Extension

The modest 2,000-step evolution effect (d=0.78, δ=0.39) grows substantially at longer timescales (Figure S8). At 10,000 steps (n=30), d=1.42, δ=0.66 ($p < 0.001$); normal populations reach $\bar{x}$=446.5 vs. 371.7 for no-evolution (Δ=−16.8%). Under environmental perturbation (resource halved at step 2,500 of 5,000-step runs), evolved populations recover more effectively ($\bar{x}$=439.5 vs. 413.5, d=0.86, δ=0.50, $p < 0.001$).

Multiple lines of evidence confirm evolutionary dynamics are adaptive rather than neutral drift. Analytical heritability is high ($h^2 \approx 0.96$). Genome drift trajectories diverge monotonically over 10,000 steps (Figure S12A; Cliff's δ=1.00, $p < 0.001$), with evolved populations accumulating drift at $\bar{d}_{10K}$=0.087 while non-evolved remain at zero. Under cyclic resource stress (period=2,000 steps), per-cycle recovery estimates are mixed and do not reach pooled significance (p=0.341; Figure S12B), treated as supplemental.

Generation-stratified physiology. Organisms stratified by generation quartile at step 10,000 (n=15 seeds;

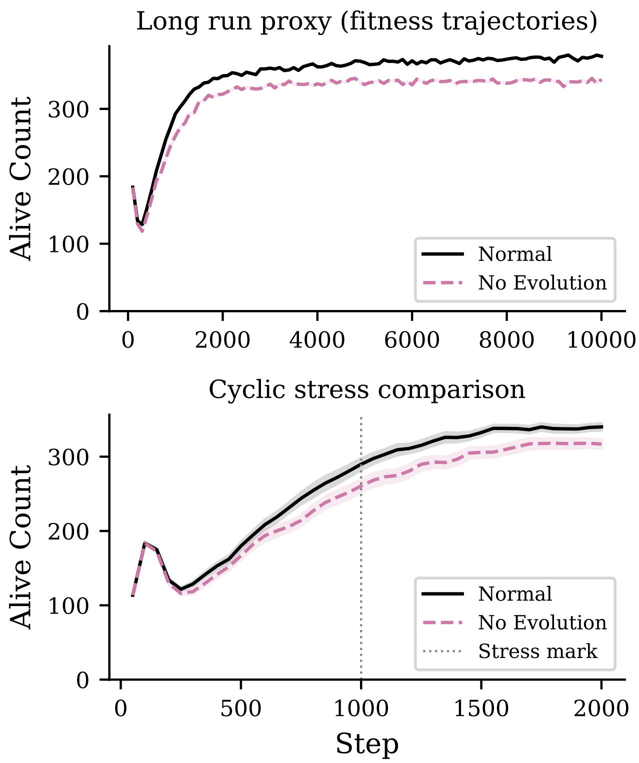

Figure S8: Evolution strengthening. Top: 10,000-step long run. Bottom: environmental shift at step 2,500 (dashed line). Lines: mean across 30 seeds; bands: ±1 SEM.

Figure S9) show significantly lower energy for high-generation organisms in both conditions (evolved: Cliff's δ=−1.00, p=3×10^{-6}; no-evolution: δ=−0.93, p=1.5×10^{-5}), reflecting reproduction cost rather than accumulated selective advantage. Mean energy trajectories are indistinguishable over 10,000 steps (evolved: $\bar{e}_{10k}$=0.885; no-evolution: 0.890); primary evidence for directional adaptation remains genome-drift divergence (δ=1.00, $p < 0.001$).

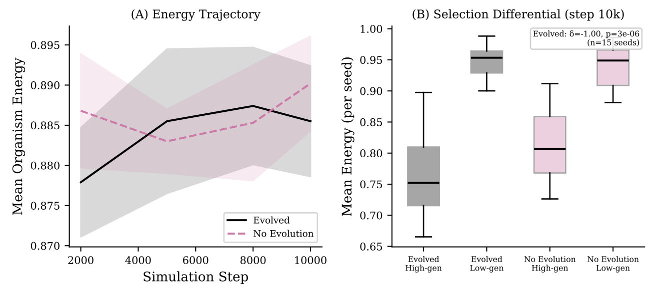

Figure S9: Generation-stratified physiology (n=15 seeds, 10,000 steps). (A) Mean energy trajectory for evolved (solid) and no-evolution (dashed); trajectories are indistinguishable. (B) Per-seed high-generation vs. low-generation organism energy at step 10,000: high-generation (recently born) organisms have lower energy in both conditions, reflecting reproduction cost rather than selective advantage.

Crossover and neutral-drift control. To distinguish genuine adaptation from neutral drift, we compare five conditions over 10,000 steps: (1) segment-wise crossover (two-parent reproduction preserving genome segment coherence); (2) no crossover (single-parent, baseline); (3) neutral drift (randomized parent selection, mutation active, removing fitness-based selection); (4) no evolution (clonal reproduction); (5) uniform crossover (sensitivity check). Segment-wise crossover is preferred over uniform because the 256-float genome contains 7 heterogeneous functional segments (NN weights, metabolic network, developmental program, etc.); uniform crossover breaks functional linkage within segments, potentially confounding results. The neutral-drift control is critical: unlike the hard "no-evolution" ablation (which removes mutation entirely), neutral drift preserves mutation but randomizes selection, isolating the contribution of fitness-based parent selection to population dynamics.

Death cause analysis. Per-step death cause tracking classifies organism deaths into three categories: boundary collapse, energy depletion, and age limit. Across all ablation conditions, boundary collapse is the univer-

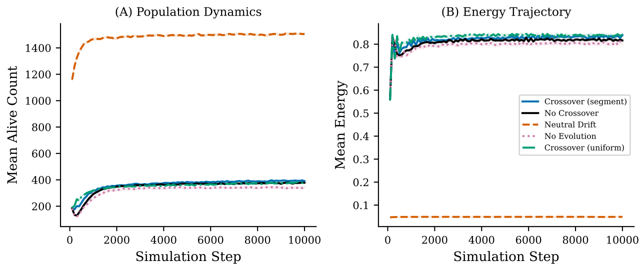

Figure S10: Fitness trajectory comparison across evolution conditions (10,000 steps, n=30 seeds). (A) Population dynamics (mean alive count ±1 SEM). (B) Energy trajectory.

sal terminal failure mode (Figure S11): boundary integrity ($b < 0.1$) consistently reaches the lethal threshold before energy reaches zero ($e \leq 0.0$), and no age limit is imposed in the default configuration. This confirms that boundary disintegration is the common final pathway through which all criterion removals ultimately cause death, consistent with the convergent failure attractor described in the Discussion.

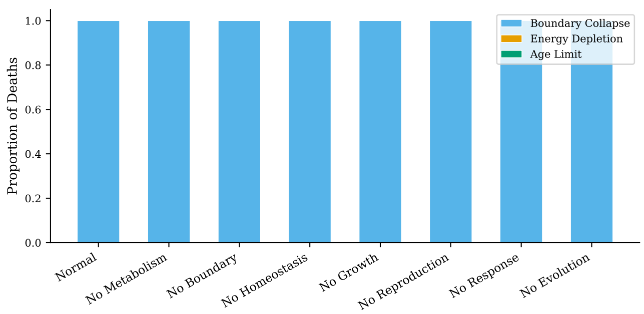

Figure S11: Death cause distribution across ablation conditions. Boundary collapse is the universal terminal cause: all criterion removals ultimately manifest as boundary disintegration ($b < 0.1$) before energy depletion or age limits are reached.

Homeostatic regulation. Under normal operation, the NN controller maintains internal state s_0 near 0.99; disabling homeostasis causes monotonic decay ($h_{\text{decay}} = 0.001$/step) with high inter-organism variance, confirming active regulation rather than static parameterization.

S7 Full Statistical Results

Table S4 reports uncorrected and Holm-Bonferroni corrected p-values, Cliff's δ with percentile bootstrap 95% CIs (2,000 resamples), and rank within the ablation effect hierarchy for all seven single-criterion ablations.

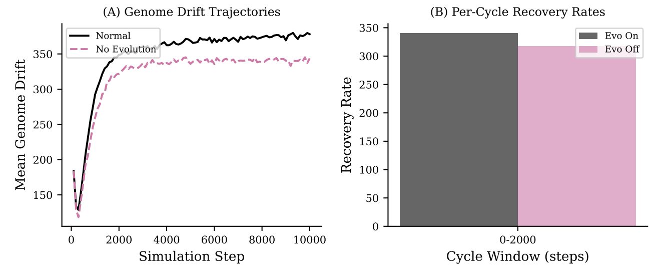

Figure S12: Evolution evidence. (A) Genome drift trajectories over 10,000 steps: evolved populations (black) accumulate genetic change linearly while non-evolved populations (purple) remain at zero (δ=1.00). (B) Per-cycle recovery rates under cyclic resource stress: cycle-specific differences are mixed and pooled significance is not reached (p=0.341).

Table S4: Full statistical results for single-criterion ablations (n=30 per condition). p_{raw}: one-sided Mann-Whitney U; p_{corr}: Holm-Bonferroni corrected (m=7, α=0.05); δ: Cliff's delta [bootstrap 95% CI, 2,000 resamples]; Rank: by $|\Delta\%|$.

Condition	$\Delta\%$	p_{raw}	p_{corr}	δ [95% CI]	Rank
No Reprod.	−91.1	<1e-10	<.001	1.00 [1.00, 1.00]	1
No Response	−88.5	<1e-10	<.001	1.00 [1.00, 1.00]	2
No Metab.	−86.8	<1e-10	<.001	1.00 [1.00, 1.00]	3
No Homeo.	−51.5	<1e-10	<.001	1.00 [1.00, 1.00]	4
No Growth	−45.9	<1e-10	<.001	1.00 [1.00, 1.00]	5
No Boundary	−35.0	<1e-9	<.001	0.99 [0.98, 1.00]	6
No Evol.	−7.8	6.8e-4	.0047	0.39 [0.11, 0.64]	7

S8 Failure Mode Analysis

To address which internal variables collapse first under each ablation, we perform a systematic failure mode analysis on existing ablation data (20 seeds per condition). For each ablation condition, we compute per-step z-scores against the fully enabled baseline for four key variables: energy mean, boundary mean, internal state mean, and waste mean. A "first break" is defined as the first timestep where $z < -2$ is sustained for ≥ 3 consecutive samples (60 steps). The cascade ordering reports which variable breaks first, second, etc.

For the essential triad: Metabolism ablation shows energy collapse as the first break (median break step 20), with waste and internal state collapsing simultaneously (step 20); boundary never reaches the $z < -2$ threshold, indicating that organisms die before boundary integrity degrades. Response ablation shows the same pattern: energy breaks first (step 20), followed by waste and internal state (step 20), then boundary decay (step 100). Reproduction ablation shows a distinct signature: only waste and internal state break (step 20), while energy and boundary remain above the $z < -2$ threshold; the un-replaced population maintains individual homeostasis but the system fails through demographic attrition.

See Figure S13 for normalized time-series with break markers.

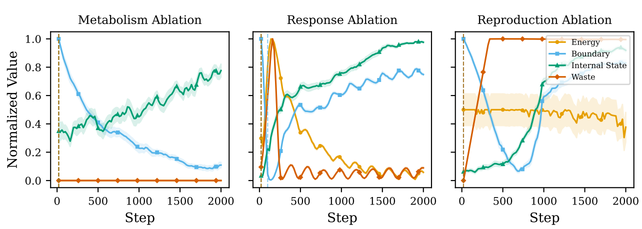

Figure S13: Failure mode analysis for the essential triad (metabolism, response, reproduction). Normalized time-series of energy (blue), boundary (orange), and internal state (green) under each ablation, with ± 1 SEM bands. Vertical dashed lines mark the median "first break" step per metric ($z < -2$ sustained for ≥ 60 steps). Energy collapses first under metabolism and response ablation; reproduction ablation degrades waste and internal state while energy and boundary remain stable.

S9 Fitness Landscape Evidence

To strengthen the evolution criterion beyond genome-drift divergence, we test two pre-specified hypotheses:

H1 (Directional selection): Mean energy of surviving organisms increases across generation cohorts in evolved populations but not in clonal (no-evolution) controls. We test this using the Jonckheere-Terpstra trend test (one-sided) on generation cohorts (0–4, 5–9, 10–14, 15+).

H2 (Heritability): Parent-offspring energy correlation is positive and significantly greater than zero. We compute the parent-offspring regression slope with bootstrap 95% CI (2,000 resamples) using lineage event linkage.

Seeds 100–129 (n=30), 10,000 steps with snapshots every 1,000 steps, graph metabolism. Two conditions: evolved (normal) vs. clonal control (evolution disabled).

Results. H1 (directional selection): The Jonckheere-Terpstra test does not reject the null ($J = 1.62 \times 10^9$, $p = 0.944$); cohort mean energies are flat across generations (0.81 for all four cohorts). The clonal control is similarly flat ($p > 0.99$). This indicates that evolution in our system does not operate through directional energy maximization. H2 (heritability): The parent-offspring regression slope is $\hat{b} = 0.747$, 95% bootstrap CI $[0.730, 0.763]$ ($n = 68{,}497$ pairs); seed-level cluster-robust CI $[0.707, 0.785]$ ($n_{\text{seeds}} = 30$), both excluding zero, indicating strong heritability of the energy phenotype. Cohen's $d = 0.058$ (evolved $n = 11{,}340$ vs. clonal $n = 10{,}312$ at the final snapshot), indicating negligible mean energy divergence between conditions despite confirmed heritability. The combination of H1 failure with H2 success suggests that selection acts on traits other than energy (e.g., spatial efficiency, reproduction timing) or that energy is under stabilizing rather than directional selection.

See Figure S14 for the 2-panel summary.

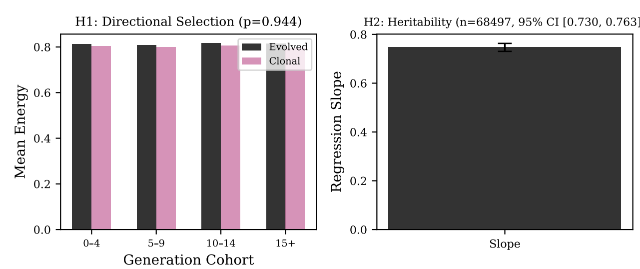

Figure S14: Fitness landscape evidence. Left: mean energy by generation cohort (evolved vs. clonal control). Right: parent-offspring regression slope with 95% bootstrap CI.

S10 Sensitivity and Robustness

Threshold sensitivity. To test whether the ablation effect hierarchy is robust to death threshold choices, we vary death_boundary_threshold over $\{0.05, 0.1, 0.15, 0.2\}$ and death_energy_threshold over $\{0.0, 0.05, 0.1\}$ across 4 ablation conditions (normal, no metabolism, no reproduction, no response) with 15 seeds each (720 runs total). We assess rank stability via Spearman correlation of each threshold combo's Δ% ranking against the default-threshold ranking (bt=0.1, et=0.0).

Results. 9 of 11 non-reference threshold combinations yield Spearman $\rho = 1.0$ with the default ranking, confirming perfect rank preservation. Two combinations (bt=0.05 with et=0.05 and et=0.1) yield $\rho = 0.5$: at these thresholds, the metabolism and response Δ% values converge (−88.3% vs. −88.3% and −88.9%, respectively), producing a near-tie that swaps their ranks. Across all 12 combinations, metabolism ablation causes Δ% ≈ −88.3 to −88.7%, reproduction ablation ≈ −91.8 to −92.2%, and response ablation ≈ −85.1 to −88.9%. Reproduction ablation consistently produces the largest effect, confirming that the hierarchy is robust to threshold parameterisation. See Figure S15.

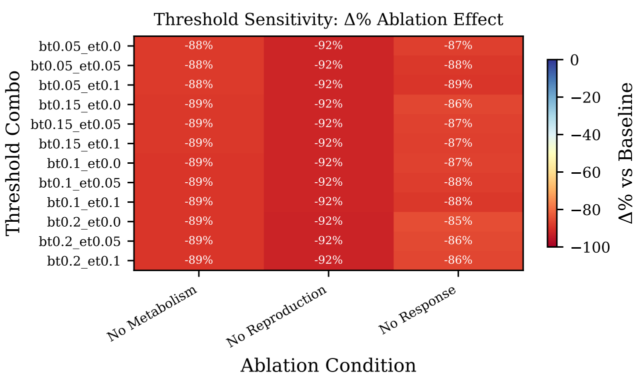

Figure S15: Threshold sensitivity heatmap: Δ% ablation effect for each threshold combination × ablation condition. Rank ordering of effects is assessed via Spearman correlation with the default-threshold ranking.

Boundary topology. To test whether findings generalize beyond the toroidal environment, we compare the ablation effect hierarchy under toroidal (wrap-around) and bounded (wall) topologies across 4 conditions with 30 seeds each (240 runs total).

Results. The Δ% rank ordering is perfectly preserved across topologies (Spearman $\rho = 1.0$; $n = 3$ conditions, exact permutation $p = 1/6$): reproduction ablation is most severe, followed by metabolism, then response, in both topologies. Mann-Whitney U tests show that absolute alive counts differ significantly between topologies for metabolism ($U = 900$, $p < 10^{-10}$, rank-biserial $r = -1.0$) and reproduction ($U = 900$, $p < 10^{-10}$, $r = -1.0$), but not for response ($U = 358$, $p = 0.175$, $r = 0.20$) or the normal baseline ($U = 533$, $p = 0.223$, $r = -0.18$). Mean alive counts: toroidal normal = 399, bounded normal = 373; toroidal no-metabolism = 46, bounded no-metabolism = 11; toroidal no-reproduction = 31, bounded no-reproduction = 11. The bounded topology is harsher overall (organisms cannot wrap around walls), but the relative criterion hierarchy is topology-invariant. See Figure S16.

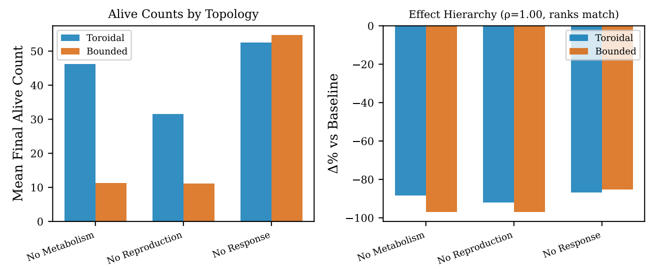

Figure S16: Boundary topology comparison. Left: mean final alive count under toroidal vs. bounded topologies for each ablation condition. Right: Δ% effect hierarchy comparison with Spearman rank correlation.

S11 Graded and Cyclic Experiments

Graded metabolic ablation. We sweep the metabolism efficiency multiplier over $\{1.0, 0.75, 0.5, 0.25, 0.0\}$ (n=30 per level, 1,000 steps, graph metabolism). Figure S17 (left) shows a monotonic dose-response: population viability decreases proportionally with metabolic efficiency (Jonckheere-Terpstra trend test, $p < 0.001$). This confirms that metabolic contribution is quantitatively proportional, not merely a binary switch; partial impairment produces proportional degradation, consistent with genuine functional analogy.

Cyclic environment. Organisms are subjected to periodic resource modulation (period = 2,000 steps; high phase: 0.01, low phase: 0.005 resource per cell) over 10,000 steps (n=30). Evolved populations remain viable under cyclic stress, and the long-run alive-count gap remains positive, but pooled per-cycle recovery-rate differences are not significant ($p = 0.341$; Figure S17, right).

S12 NN Controller Sensitivity

To rule out that ablation results depend on a specific neural network architecture, we sweep the hidden layer size over $\{8, 16, 32\}$ neurons (n=30 seeds per size, 2,000 steps). A Kruskal-Wallis test shows no significant effect of hidden size on final population count ($H = 4.61$, $p = 0.100$): hidden=8 ($\bar{x} = 348.2 \pm 35.5$), hidden=16 ($\bar{x} = 348.9 \pm 29.8$), hidden=32 ($\bar{x} = 339.4 \pm 19.4$). Pairwise Mann-Whitney comparisons are also non-significant (8 vs. 16: $U = 492$, $p = 0.544$; 16 vs. 32: $U = 554$,

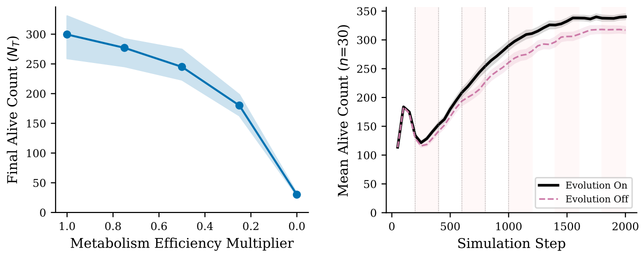

Figure S17: Left: graded metabolic ablation dose-response (median ± IQR, n=30 per level; $p < 0.001$, Jonckheere-Terpstra). Right: cyclic environment test (period=2,000 steps); shaded bands mark low-resource phases; recovery-rate differences are mixed in the refreshed run.

$p = 0.128$). This confirms that the ablation effect hierarchy is robust to NN capacity. This sweep tests baseline performance sensitivity to NN capacity; a crossed design (NN size × ablation condition) verifying hierarchy preservation across architectures remains future work.

S13 Counter/Toy Engine Comparison

To verify that criterion effects are not artifacts of the graph-based metabolic engine, we compare ablation effects across two alternative engines: a counter-based system (metabolism as a simple integer counter) and a toy engine (minimal state machine). Both engines reproduce the same effect hierarchy with all seven single-criterion ablations significant (Holm-Bonferroni corrected, all $p_{\text{adj}} < 0.001$; Cliff's $\delta = 1.00$ for 6 of 7 criteria). Evolution ablation shows the weakest effect in both engines (Counter: $\delta = 0.55$, $p_{\text{adj}} = 0.0003$; Toy: $\delta = 0.61$, $p_{\text{adj}} < 0.001$), consistent with the graph engine results. This cross-engine replication strengthens the claim that criterion necessity is a system-level property, not an artifact of the metabolic implementation.

S14 Growth/Reproduction Separability

To address the concern that growth functions primarily through reproduction gating (the maturity requirement), we compare four conditions (n=30): normal ($\bar{x} = 348.9$), no growth ($\bar{x} = 188.8$), bypass maturity gate ($\bar{x} = 419.6$), and no growth with bypassed gate ($\bar{x} = 271.4$). All pairwise comparisons are significant (Mann-Whitney U, all $p < 0.001$). The growth effect persists even when the maturity gate is bypassed (bypass vs. no-growth-bypass: $\Delta = +148.2$, $U = 900$, $p < 0.001$), confirming that growth contributes independently through boundary repair efficiency, sensing range, and metabolic throughput, not solely through the reproduction maturity gate.